Chronic Kidney Disease

What is chronic kidney disease?

Chronic kidney disease (CKD and chronic renal disease) means that there's damage to your kidneys and they aren't working as well as they should. Your kidneys are like a filter in your body — filtering out wastes, toxins and extra water from your blood. They also help with other functions like bone and red blood cell health. When your kidneys begin to lose their function, they can't filter waste, which means the waste builds up in your blood.

Kidney disease is called "chronic" because kidney function slowly decreases over time. CKD can lead to kidney failure, which is also called end-stage kidney disease. Not everyone with CKD will develop kidney failure, but the disease will often worsen without treatment. There's no cure for chronic kidney disease. But there are steps you can take to slow kidney damage. Treatments like dialysis and transplantation are options for kidney failure (end-stage kidney disease).

What do your kidneys do?

You have two kidneys. They're bean-shaped organs that are located toward your back, on either side of your spine, just underneath your rib cage. Each kidney is about the size of your fist.

Your kidneys have many jobs, but their main job is to clean your blood, getting rid of toxins, waste and excess water as urine (pee). Your kidneys also balance the amount of electrolytes (such as salt and potassium) and minerals in your body, make hormones that control blood pressure, make red blood cells and keep your bones strong. If your kidneys are damaged and don't work as they should, wastes can build up in your blood and make you sick.

What are the 5 stages of chronic kidney disease?

There are five stages of chronic kidney disease. The stages are based on how well your kidneys are able to filter out waste from your blood. Blood and urine tests determine which stage of CKD you're in.

The stages range from very mild (stage 1) to kidney failure (stage 5). Healthcare providers determine the stage of your kidney function according to the glomerular filtration rate (GFR). Your GFR is a number based on the amount of creatinine, a waste product, found in your blood.

Stage
Stage 1
GFR (mL/min)

Stage
90 and higher
What It Means
Your kidneys are working well but you have signs of mild kidney damage.
Stage 2
GFR (mL/min)
60 to 89
What It Means
Your kidneys are working well but you have more signs of mild kidney damage.
Stage 3a
GFR (mL/min)
45 to 59
What It Means
Your kidneys aren’t working as well as they should and show mild to moderate damage. This is the most common stage. You may notice symptoms at this stage.
Stage 3b
GFR (mL/min)
30 to 44
What It Means
Your kidneys show moderate damage and don’t work as well as they should. With the right treatment, many people can stay in this stage and never advance to stage 4.
Stage 4
GFR (mL/min)
15 to 29
What It Means
You have very poor kidney function; your kidneys are severely damaged and close to not working.

Stage
Stage 5
GFR (mL/min)
Less than 15
What It Means
Your kidneys are very close to failing or have stopped working. You may need kidney dialysis or a kidney transplant at this stage.

How common is this condition?

About 15% of adults in the United States have chronic kidney disease. Some 37 million people in the U.S. are living with chronic kidney disease.

Symptoms and Causes

What are the symptoms of chronic kidney disease?

In the early stages of kidney disease, you usually don't have noticeable symptoms. As the disease worsens, symptoms may include:

- A need to pee more often.
- Tiredness, weakness, low energy level.
- Loss of appetite.
- Swelling of your hands, feet and ankles.
- Shortness of breath.
- Foamy or bubbly pee.
- Puffy eyes.
- Dry and itchy skin.
- Trouble concentrating.
- Trouble sleeping.
- Numbness.
- Nausea or vomiting.
- Muscle cramps.
- High blood pressure.

- Darkening of your skin.

Keep in mind that it can take years for waste to build up in your blood and cause symptoms.

What does it feel like when something is wrong with your kidneys?

You typically don't have any signs of kidney disease, especially in its early stages. Once you begin having symptoms, the first sign something is wrong may involve swelling in your hands and feet, itchy skin or needing to pee more often. Since symptoms vary, it's best to call your healthcare provider if you believe there's something wrong.

What are common causes of kidney disease?

Kidney diseases happen when your kidneys are damaged and can't filter your blood. With chronic kidney disease, the damage tends to happen over the course of several years.

High blood pressure (hypertension) and diabetes are the two most common causes of chronic kidney disease. Other causes and conditions that affect kidney function and can cause chronic kidney disease include:

- **Glomerulonephritis**. This type of kidney disease involves damage to the glomeruli, which are the filtering units inside your kidneys.
- **Polycystic kidney disease**. This is a genetic disorder that causes many fluid-filled cysts to grow in your kidneys, reducing the ability of your kidneys to function.
- **Membranous nephropathy**. This is a disorder where your body's immune system attacks the waste-filtering membranes in your kidney.
- **Obstructions of the urinary tract** from kidney stones, an enlarged prostate or cancer.
- **Vesicoureteral reflux**. This is a condition in which pee flows backward back up your ureters to your kidneys.
- **Nephrotic syndrome**. This is a collection of symptoms that indicate kidney damage.
- **Recurrent kidney infection (pyelonephritis)**.
- **Diabetes-related nephropathy**. This is damage or dysfunction of one or more nerves, caused by diabetes.
- **Lupus and other immune system diseases that cause kidney problems**, including polyarteritis nodosa, sarcoidosis, Goodpasture syndrome and Henoch-Schönlein purpura.

Is kidney disease hereditary?

Yes, kidney disease can run in biological families. Risk factors for CKD, like diabetes, also tend to run in families.

Who is at risk for chronic kidney disease?

Anyone can get chronic kidney disease. You're more at risk for chronic kidney disease if you:

- Have diabetes.
- Have high blood pressure.
- Have heart disease.
- Have a family history of kidney disease.
- Have abnormal kidney structure or size.
- Are over 60 years old.
- Have a long history of taking NSAID (nonsteroidal anti-inflammatory drugs) pain relievers. This includes over-the-counter (OTC) products and some prescription pain relievers.

What are the complications of chronic kidney disease?

Some of the complications of chronic kidney disease include:

- A low red blood cell count (anemia).
- Weak and brittle bones.
- Gout.
- Metabolic acidosis. This is a chemical imbalance (acid-base) in your blood caused by a decrease in kidney function.
- High blood pressure.
- Heart disease and blood vessel disease, including increased risk of stroke and heart attack.
- Nerve damage.
- High potassium (hyperkalemia), which affects your heart's ability to function correctly.
- High phosphorus (hyperphosphatemia).
- High risk of infection due to a weak immune system.
- Fluid buildup, leading to swelling in your feet, ankles and hands.

Diagnosis and Tests

How is kidney disease diagnosed?

First, your healthcare provider will take your medical history, conduct a physical exam, ask about any medications you're currently taking and ask about any symptoms you've noticed.

Your healthcare provider will order blood and urine tests to check kidney function.

Specifically, the blood tests will check:

- Your glomerular filtration rate (GFR). This describes how efficiently your kidneys are filtering blood — how many milliliters per minute your kidneys are filtering. Your GFR is used to determine the stage of your kidney disease.
- Your serum creatinine level, which tells how well your kidneys are removing this waste product. Creatinine is a waste product from muscle metabolism and is normally excreted in your pee. A high creatinine level in your blood means that your kidneys aren't functioning well enough to get rid of it in your pee.

Urine tests will look for protein (albumin) and blood in your pee. Well-functioning kidneys shouldn't contain either.

Other tests may include imaging tests to look for problems with the size and structure of your kidneys — such as ultrasound, magnetic resonance imaging (MRI) and/or computerized tomography (CT) scans. Your healthcare provider may also order a kidney biopsy to check for a specific type of kidney disease or to determine the amount of kidney damage.

Management and Treatment

How is chronic kidney disease treated?

There's no cure for chronic kidney disease (CKD), but steps can be taken to preserve your kidney function so they work as long as possible. If you have reduced kidney function:

- Make and keep your regular healthcare provider/nephrologist (kidney specialist) visits. These providers monitor your kidney health.
- Manage your blood glucose (sugar) if you have diabetes.
- Avoid taking painkillers and other medications that may make your kidney disease worse.
- Manage your blood pressure levels.
- Follow a kidney-friendly diet. Dietary changes may include limiting protein, eating foods that reduce blood cholesterol levels and limiting sodium (salt) and potassium intake.
- Don't smoke.
- Exercise/be active on most days of the week.
- Stay at a weight that's healthy for you.

Medications for kidney disease

Depending on the cause of your kidney disease, you may be prescribed one or more medications. Medications your nephrologist may prescribe include:

- An angiotensin-converting enzyme (ACE) inhibitor or an angiotensin receptor blocker (ARB) to lower your blood pressure.
- Phosphate binder if your kidneys can't eliminate phosphate.
- A diuretic to help your body eliminate extra fluid.
- Medications to lower cholesterol levels.
- Erythropoietin to build red blood cells if you're anemic.
- Vitamin D and calcitriol to prevent bone loss.

What is kidney dialysis?

Because there's no cure for CKD, if you're in end-stage kidney disease, you and your healthcare team must consider additional options. Complete kidney failure will result in death if it's left untreated. Options for end-stage kidney disease include dialysis and kidney transplantation.

Dialysis is a procedure that uses machines to remove waste products from your body when your kidneys are no longer able to perform this function. There are two major types of dialysis:

- **Hemodialysis:** With hemodialysis, your blood is circulated through a machine that removes waste products, excess water and excess salt. The blood is then returned to your body. Hemodialysis requires four-hour treatments, three times a week.
- **Peritoneal dialysis:** In peritoneal dialysis, a dialysis solution is placed directly into your abdomen through a catheter. The solution absorbs waste and then is removed via the same catheter. Fresh solution is added to continue the process of cleaning. You can perform this type of dialysis yourself. There are two types of peritoneal dialysis: continuous ambulatory peritoneal dialysis (CAPD), which involves a change in dialysis solution four times a day; and continuous cycling peritoneal dialysis (CCPD). CCPD uses a machine to automatically fill, remove and refill the fluid during the nighttime.

What is a kidney transplant?

Kidney transplantation involves replacing an unhealthy kidney with a healthy kidney. Kidneys for transplantation come from two sources: living donors and deceased donors. Living donors are usually family members, partners or friends. A living kidney donor is possible because a person can live well with one healthy kidney.

Deceased donor kidneys usually come from people who are organ donors. All donors are carefully screened to make sure there's a suitable match and to prevent any transmissible diseases or other complications.

On average, people wait about three to five years for a kidney from a deceased donor. It's usually quicker to receive a kidney from a living donor.

Outlook / Prognosis

What can I expect if I have kidney disease?

If you have kidney disease, you can still live a productive home and work life and enjoy time with your family and friends. To have the best outcome possible, it's important for you to become an active member of your treatment team.

Early detection and appropriate treatment are important in slowing the disease progression, with the goal of preventing or delaying kidney failure. You'll need to keep your medical appointments, take your medications as prescribed, stick to a nutritious diet and monitor your blood pressure and blood sugar.

Prevention

Can kidney disease be prevented?

Seeing your healthcare provider on a regular basis throughout your life is a good start for preventing kidney disease. About 1 in every 3 people in the United States is at risk for kidney disease. People at high risk may have regular tests to check for CKD so it's detected as early as possible. Some other things you can do to prevent CKD are:

- Manage your high blood pressure.
- Manage your blood sugar if you have diabetes.
- Eat a well-balanced diet.
- Don't smoke or use tobacco.
- Be active for 30 minutes at least five days a week.
- Maintain a healthy weight.
- Take nonprescription pain relievers only as directed. Taking more than directed can damage your kidneys.
- Limit alcohol-containing beverages.

Living With

How long can someone live with chronic kidney disease?

While CKD can lead to death, many people with the condition live long and happy lives after diagnosis. Most people who seek treatment for kidney disease and manage their condition never progress to kidney failure or death. That's why it's important to attend all your checkups and work with your healthcare provider on a treatment plan.

The leading cause of death in people with CKD is actually heart disease, a complication of CKD. Managing other health conditions that negatively impact your kidneys is also key to maintaining your kidney function.

When should I see my healthcare provider?

Early detection can help prevent kidney disease from worsening to kidney failure. Work with your healthcare provider to manage conditions that are known to cause kidney disease. These include diabetes, high blood pressure and other diseases that affect your kidneys.

Since kidney disease often doesn't cause symptoms in the early stages, the best thing you can do is work with your provider to understand your risk and attend all annual or scheduled visits with your provider.

Most people don't have symptoms until CKD is severe. Contact your provider if you:

- Feel more tired than usual.
- Don't feel like eating.
- Start peeing more than usual.
- Have trouble sleeping or focusing.
- Have muscle cramps, itchy skin or swollen feet and ankles.

Additional Common Questions

How do you know if your kidneys are struggling?

You may not know your kidneys are struggling. Most people don't have symptoms of kidney disease in the early stages. That's why it's important to attend annual wellness exams with your primary care provider to manage chronic conditions like diabetes or high blood pressure that can lead to kidney disease.

What foods are bad for kidneys?

In people with healthy kidneys, there aren't necessarily bad foods or foods that hurt your kidneys. But, if you have CKD, your healthcare provider may recommend a kidney-friendly diet. Elements of a kidney-friendly diet may include:

- **Avoiding foods that are high in salt**. This also helps control blood pressure.

- **Eating the right amount of protein**. Protein creates more waste than other food groups. So, since your kidneys remove waste, lowering protein can help preserve their function.
- **Eating heart-healthy foods**.
- **Eating foods low in phosphorus**. This includes fresh fruits and vegetables and whole grains. Foods like dairy and beans are high in phosphorus.
- **Avoiding foods high in potassium** like bananas, oranges and potatoes.

Since following a kidney-friendly diet is hard to understand and to do, it's always a good idea to consult a dietitian as part of your treatment plan. They can help make sure you're eating the right types of food if you have chronic kidney disease.

What color is urine when your kidneys are failing?

Your pee shouldn't change color, but may be foamy or frothy, which means there's excess protein in your pee. Excess protein means your kidneys aren't filtering toxins from your body.